

Ayush Kumar Verma

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Professional Summary

MCA (AI & ML) student at UPES with foundations in software development, machine learning, and data analysis. Proficient in Java, Python, and SQL, with hands-on experience building software and AI-enabled applications. Currently developing LeanAI, an LLM context and token optimization system designed to minimize unnecessary repository context through intelligent context selection, token budgeting, and model routing. Interested in applying Generative AI and Agentic AI techniques to practical software engineering problems.

Education

Master of Computer Applications (AI & ML)	Jul 2025 – Jul 2027
UPES	CGPA: 8.02/10
Bachelor of Computer Applications	2022 – 2025
SRHU	CGPA: 7.3
Class XII (CBSE)	77.8%
Class X (CBSE)	70.2%

Technical Skills

Programming Languages: Java, Python, C, JavaScript, SQL

Generative & Agentic AI: Large Language Models (LLMs), Multi-Agent Systems, Context Engineering, Prompt Engineering, Token Optimization

Machine Learning: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, Matplotlib

Web / Development: React, HTML5, CSS3, Javascript.

Tools & Practices: Git, GitHub, VS Code, Jupyter Notebook, Version Control, Documentation

Projects

LeanAI – LLM Context & Token Optimization System

- Currently developing an LLM context and token optimization system that analyzes software repositories and selects task-relevant code, reducing unnecessary input context instead of sending entire codebases to LLMs.
- Engineering a Rust-based context retrieval and bundling pipeline using relevance-based file selection, token estimation, and Git-aware signals to optimize repository context supplied to LLMs.
- Designing a dynamic context-selection and token-budgeting workflow to determine relevant repository information for coding tasks and support efficient model routing.
- Prototyping agentic AI and multi-agent workflows for planning, analysis, coding, review, and testing within a Tauri desktop application with a React and TypeScript interface.

Travillo – Travel Assistance Platform

- Built a travel planning platform with personalized trip-planning features, designing responsive, client-focused pages using HTML, CSS, and JavaScript.
- Integrated the Mapbox API to enable interactive, location-based functionality for exploring destinations and planning routes.
- Implemented community discussion features to support user collaboration and travel experience sharing.

Certifications

- Honeywell Cybersecurity Training – ICT Academy (Grade A), covering data security and privacy fundamentals
- Certificate in Search Engine Optimization (SEO)